Harsharandeep Dhillon

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EDUCATION

University of California, Berkeley **Berkeley, CA**
Bachelors in Applied Mathematics with Data Science Concentration [GPA - 3.7] *May 2027*
Availability: Graduating May 2027, available for 2027 start
Organizations/Awards: AI/ML fellow alum at BTT Cornell Tech, Academic Honors List, Women In Tech Club Member

WORK EXPERIENCE

Zoox - Amazon Robotaxi **Foster City, CA**
AI Software Engineer (Contract) *July 2026 - Present*

- Designed and deployed MCP server integrations connecting AI agents to production supply chain, quality and reliability (SCQR) data including supplier scorecards, non conformance reports, and audit findings, using data and metrics to evaluate retrieval quality, eliminating manual lookup across thousands of internal data sources and cutting review time by roughly 60%
- Built and shipped a production retrieval augmented generation (RAG) pipeline, containerized with Docker and deployed on AWS, that applies information retrieval techniques to turn raw reliability and field quality records into a ranked, agent queryable system, adopted into the team's daily workflow within two weeks
- Built a live ETL pipeline connecting Databricks to production supply chain database systems through internal APIs, designing data models that automate the flow from supplier part shipments through fleet wide part cost reporting across thousands of internal data sources, replacing a manual, disconnected process
- Leading end to end development of full stack internal tools across multiple SCQR teams, gathering data requirements directly from engineering and business stakeholders, building the underlying infrastructure, and shipping applications from API and MCP integration through containerization with Docker and deployment to Kubernetes
- Provisioned and managed AWS infrastructure including S3 and EC2 to support production data pipelines and containerized applications, and engineered large scale data streaming and processing with Spark and Hadoop across structured and non relational data sources for fleet and geospatial telemetry

Multibeam Corporation - Semiconductor Systems **Sunnyvale, CA**
Data Engineering Intern *Jan 2026 - June 2026*

- Developed a Python based automation pipeline for large scale SEM image validation and classification, applying statistical analysis and machine learning to reduce manual review workload by 80% and improve defect detection accuracy
- Queried and validated large scale production datasets using SQL and custom Python tooling against relational database systems, implementing image processing algorithms including contrast normalization, edge detection, and noise filtering to improve model accuracy and pipeline efficiency
- Designed reusable Python modules and automated statistical data validation workflows with Linux based scripting, improving data quality, pipeline reliability, and data governance while reducing manual reporting effort by 40%
- Collaborated cross functionally with hardware and software teams to troubleshoot lithography system performance, investigate root causes through data driven analysis, and improve operational reliability

KPMG x Break through tech
Machine learning fellow *May 2025 - December 2025*

- Conducted LLM evaluation and benchmarking using the Hugging Face Energy Dataset (40K+ model runs) across multiple open source and proprietary models, building an evaluation framework to analyze trade offs between performance, energy usage, and CO2 emissions and inform measurable ROI and efficiency KPIs
- Engineered data driven AI evaluation workflows using Python (Pandas, NumPy, Scikit learn), R, and SQL based querying, designing EDA visualizations with Matplotlib and Seaborn to communicate model trade offs to technical and business stakeholders
- Built and deployed an interactive Streamlit dashboard and automated reporting tool to visualize real time AI evaluation, ROI, and emissions metrics, enabling data driven decision making for enterprise AI adoption

PROJECTS

Airbnb Price Prediction Model
Machine Learning and Data Science Project

- Built a scalable data pipeline to clean, transform, and prepare large scale Airbnb datasets for modeling, leveraging Apache Spark for distributed processing of listing and pricing data across multiple markets
- Designed and orchestrated automated workflows using Apache Airflow for dataset ingestion, preprocessing, and feature engineering, training predictive models to quantify pricing trends by location and host attributes

AI Meeting Assistant
Full Stack AI Productivity Application

- Built an AI powered meeting assistant using Next.js, TypeScript, Tailwind CSS, and OpenAI APIs with prompt engineering to analyze meeting transcripts and generate structured summaries, action items, and project features
- Engineered backend API workflows and full stack interfaces for AI driven task extraction, deploying cloud infrastructure on Vercel and AWS with attention to data handling and governance

EXTRACURRICULAR ACTIVITIES AND SKILLS

Skills: Python, R, SQL, Database Systems, Non Relational Databases, Data Warehousing, ETL, Data Modeling, PyTorch, TensorFlow, Spark, Hadoop, Databricks, AWS (S3, EC2), Docker, Kubernetes, GitHub Actions, CI/CD, Git, REST APIs, Pandas, NumPy, Scikit Learn, LLMs, RAG, Information Retrieval, Machine Learning, Apache Airflow, Streamlit, Dashboards and Reporting, Data Visualization
Course work: Data Structures and Algorithms, Machine Learning Foundations, Object Oriented Programming, Calculus, Linear Algebra